

An Unofficial Compendium of Worked Solutions to the Problems in *Geometry*

by I. M. Gelfand & T. Alekseyevskaya (Gelfand)

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Abstract

This book is an independent and unofficial collection of worked solutions to problems inspired by the book *Geometry* by I. M. Gelfand and T. Alekseyevskaya (Gelfand), first published in 2020.

The original text presents a wide range of problems in projective geometry, affine geometry, symplectic geometry, and Euclidean geometry. The book contains solved exercises and questions, but the reader is also given over 200 problems to solve.

The aim of this book is to provide clear, logically consistent solutions to all unsolved problems addressed, written in a style accessible to motivated students and self-learners.

Figures are either drawn by hand and scanned, constructed using GeoGebra, or produced directly in $\text{\LaTeX}$. Where applicable, proofs or calculations may also be checked using computer programs written in Scheme (a dialect of Lisp) or Python.

Large language models were used only as auxiliary tools, primarily for generating figures in $\text{\LaTeX}$ and for reviewing exposition.

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Issues and corrections may be reported at the [project repository](#).

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Unofficial companion.

This document is an **independent, unofficial collection of worked solutions** to problems from *Geometry* by I.M. Gelfand and T. Alekseyevskaya (Gelfand) (first published in 2020).

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The original problem statements are not reproduced here. All exposition and solutions are written independently by the author of this document.

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Project repository:

<https://github.com/deepak-venkatesh/gelfand-geometry>

For
My Daughter and Son

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Working Notes

Challenging Problems for Middle Schoolers:

Typographical Errors:

1. The standard mathematical spelling of Hermann Minkowski's name is *Minkowski* and not *Minkowsky*.

Introduction

No problems in this chapter.

Part I

Points and Lines: A Look at Projective Geometry

Chapter 1. Points and lines

Problem 1.

In a line if we place a point we divide it into 2 parts. In this case both the part are *rays* and the point is an *endpoint*.

Now if there are 2 point on the line, we still have 2 rays but the two endpoints create a line segment.

So n points on a line will give 2 rays and $(n - 1)$ line segments.

For this problem.

$$\begin{aligned} n &= 100 \\ \text{endpoints} &= 100 \\ \text{rays} &= 2 \\ \text{segments} &= 100 - 1 = 99 \end{aligned}$$

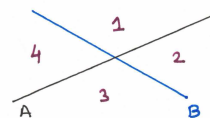
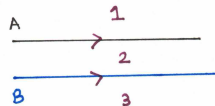
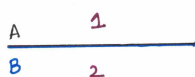


Problem 2.

Three possibilities exist:

1. The two lines A and B can perfectly coincide.
2. They can be parallel to each other.
3. They can intersect at some point.

Basis the figure below the plane can be divided into 2, 3, or 4 parts.



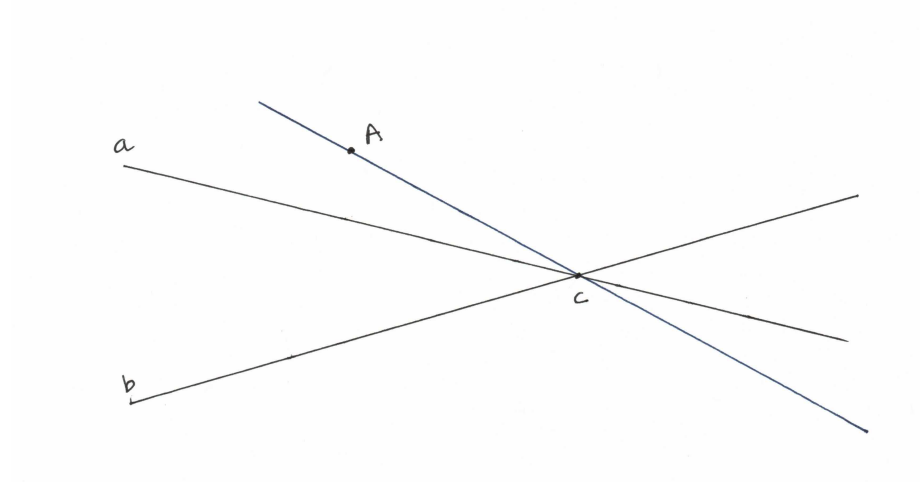
Problem 3.

The chapter explains that geometric constructions performed using the two axioms below are call constructions in *projective geometry*.

Axiom 1: Through any two points, one can draw a unique line.

Axiom 2: Any two lines either have a unique point of intersection or do not intersect (are parallel).

Basis these two axioms of projective geometry we can construct the line AC .



Chapter 2. Two lines and an angle

Chapter 3. Three lines

Chapter 4. Four lines. Quadrilaterals

Chapter 5. Five lines

Chapter 6. Projection from a point onto a line

Chapter 7. Dual configurations in projective geometry

Chapter 8. Desargues configuration

Chapter 9. Dual Desargues configuration

Chapter 10. Algebraic notation of “computer presentation” of configurations

Chapter 11. Polygons and n straight lines

Chapter 12. Convex polygons, convex hull of n points

Chapter 13. Solution of Exercise 3 with the help of a Desargues configuration

Chapter 14. Overview of Chapter I

Part II

Parallel Lines: A Look at Affine Geometry

Part I

Lines and Segments

Chapter 1. Parallel Straight Lines

Chapter 2. Operations available in Chapter II

Chapter 3. Properties of parallel lines

Chapter 4. Segments lying on parallel lines

Part II

Figures

Chapter 5. Parallelograms

Chapter 6. Triangles

Chapter 7. Trapezoids

Part III

Operations with Figures

Chapter 8. The Minkowsky addition of two figures

Chapter 9. Parallel projection

Chapter 10. Parallel translation

Chapter 11. Central symmetry on the plane

Chapter 12. Vectors

Chapter 13. Overview of Chapter II

Appendix for Chapter II

1. Why we cannot define equal segments in Chapter II
2. Parallel lines, equal segments, and the Desargues configuration
3. Arithmetic operations with segments
4. Segments and rational numbers
5. Affine coordinate systems on the plane

Part III

Area: A Look at Symplectic Geometry

Chapter 1. The area of a figure

Chapter 2. Area of a parallelogram

Chapter 3. Area of a triangle

Chapter 4. Area of a trapezoid

Chapter 5. Area of a polygon

Chapter 6. More problems on areas

Chapter 7. How to measure the area of a figure

Chapter 8. Overview of Chapter III

Part IV

Circles: A Look at Euclidean Geometry

Part I

Introduction to the Circle

Chapter 1. Operations available in Chapter IV

Chapter 2. Comparing segments

Chapter 3. Angles

Chapter 4. Operations with figures

Part II

**The Geometry of the Triangle and
Other Figures**

Chapter 5. Elements of triangle. Congruent triangles

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Part III

Circles

Chapter 16. Circles and points

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Chapter 18. Two or more circles

Chapter 19. Circles and angles

Chapter 20. A circle and a triangle

Chapter 21. Circles and polygons

Chapter 22. Circumference and arc

Chapter 23. Disks and sectors

Chapter 24. Overview of Chapter IV